

A polygenic risk score for peripheral artery disease and major adverse limb events

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Received 14 November 2024; revised 21 April 2025; accepted 21 October 2025; online publish-ahead-of-print 28 November 2025

See the editorial comment for this article 'From biobank to prediction: polygenic risk scores for peripheral artery disease', by E. Fukaya and N.J. Leeper, <https://doi.org/10.1093/eurheartj/ehaf1042>.

Abstract

Background and Aims	Large-scale genome-wide association studies have identified common genetic variants that predict the risk of peripheral artery disease (PAD). This study assessed whether a polygenic risk score (PRS) is associated with PAD and the incidence of major adverse limb events (MALE) independent of clinical risk factors in patients with established cardiometabolic disease.
Methods	A genetic analysis was performed, pooling individual patient-level data from six TIMI trials. The association of a recently validated PAD PRS with prevalent PAD and the incidence of MALE (acute limb ischaemia, chronic limb-threatening ischaemia, major amputation, or peripheral revascularization) was assessed.
Results	A total of 68 816 patients were included in this analysis, with a median follow-up of 2.6 years. Of these, 5986 (8.7%) had known PAD at baseline. After adjusting for clinical risk factors, a higher PAD PRS was independently associated with a 15% greater odds of prevalent PAD (adjusted odds ratio per 1-SD: 1.15 [95% confidence interval 1.12–1.18], $P < .0001$), a magnitude of risk as strong as established clinical risk factors. A total of 577 patients experienced MALE during follow-up. A higher PAD PRS was associated with a 30% increased risk of MALE (adjusted hazard ratio per 1-SD: 1.30 [1.19–1.42], $P < .0001$). Adding the PAD PRS to clinical risk factors resulted in a statistically significant but modest improvement in discrimination (area under the curve went from 0.651 to 0.662 $P < .0001$).
Conclusions	In a broad spectrum of patients with cardiometabolic disease, the PAD PRS is associated with an increased risk of PAD and the incidence of MALE beyond clinical risk factors; however, the improvement in discrimination was statistically significant but clinically modest.

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Structured Graphical Abstract

Key Question

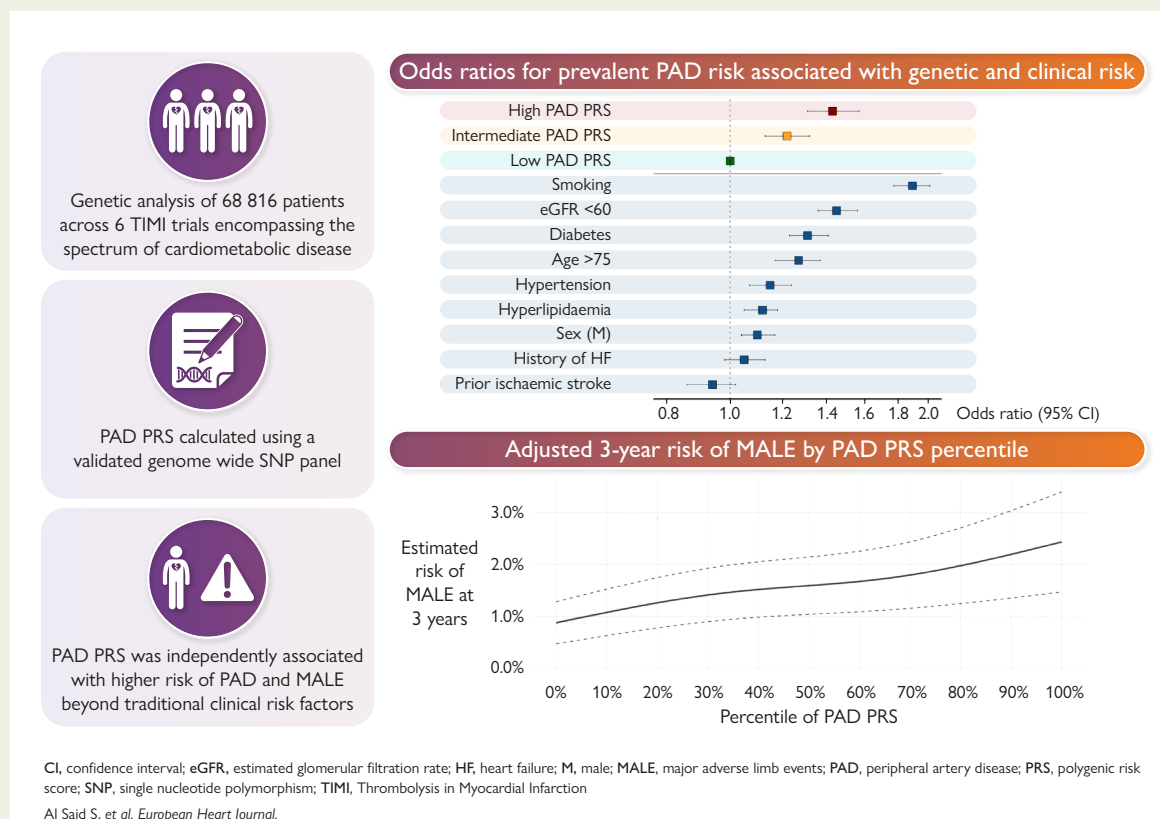
What is the role of a polygenic risk score (PRS) in predicting peripheral artery disease (PAD) and major adverse limb events (MALE) in patients with established cardiometabolic disease?

Key Finding

In ~69,000 patients from 6 TIMI trials, the PAD PRS was independently associated with PAD and MALE, beyond clinical factors.

Take Home Message

A PAD PRS is independently associated with PAD and limb events in a broad spectrum of patients with cardiometabolic disease with clinically modest improvement in discrimination.



The relationship between the polygenic risk score (PRS) for peripheral artery disease (PAD) and clinical outcomes. *Top panel*: adjusted odds ratios for prevalent PAD according to PAD PRS category and traditional clinical risk factors. *Bottom panel*: adjusted 3-year risk of major adverse limb events (MALE) across continuous PAD PRS. Abbreviations: CI, confidence interval; eGFR, estimated glomerular filtration rate; HF, heart failure; M, male; MALE, major adverse limb event; PAD, peripheral artery disease; PRS, polygenic risk score; SNP, single-nucleotide polymorphism; TIMI, thrombolysis in myocardial infarction.

Keywords

Polygenic risk score • Peripheral artery disease • PAD • Major adverse limb events • MALE

Introduction

Peripheral artery disease (PAD) remains a significant cause of cardiovascular morbidity and mortality.^{1,2} Despite the severity of this condition, PAD often goes undiagnosed and untreated until the late stages, when patients become symptomatic.^{3–6} Patients with PAD are at an increased risk for major adverse limb events (MALE), including acute limb ischaemia (ALI), chronic limb-threatening ischaemia (CLTI), major amputation, and peripheral revascularization.^{3–6} Traditional risk factors for atherosclerosis, including hypertension, diabetes, and smoking, have

been strongly linked to the development of PAD.⁷ Studies have further suggested that genetic determinants, independent of traditional risk factors, may also modify the risk for development of PAD.⁸

Large-scale genome-wide association studies (GWAS) have further clarified the genetic architecture of PAD, identifying common variants that predict the risk of PAD in several cohorts consisting of patients primarily without established cardiovascular disease.^{9–12} However, it remains unclear whether a PAD polygenic risk score (PRS) is associated with prevalent PAD, independent of clinical risk factors, in patients predominantly with established cardiometabolic disease. Additionally,

whether a PAD PRS is associated with incident MALE has not been well established.

Accordingly, we sought to (i) assess whether a validated PAD PRS is associated with prevalent PAD in a broad patient population encompassing the spectrum of cardiometabolic disease, (ii) examine whether a PAD PRS is associated with the incidence of MALE, and (iii) assess the performance of a PAD PRS in detecting the risk of events in other vascular beds, including incident myocardial infarction (MI) and ischaemic stroke.

Methods

Study design and population

We performed a genetic data analysis by pooling individual patient-level data across six clinical trials encompassing the spectrum of cardiometabolic disease: ENGAGE AF-TIMI 48 (Effective Anticoagulation with Factor Xa Next Generation in Atrial Fibrillation),¹³ SOLID-TIMI 52 (Stabilization of Plaques Using Darapladib),¹⁴ SAVOR-TIMI 53 (Saxagliptin Assessment of Vascular Outcomes Recorded in Patients with Diabetes Mellitus),¹⁵ PEGASUS-TIMI 54 (Prevention of Cardiovascular Events in Patients with Prior Heart Attack Using Ticagrelor Compared to Placebo on a Background of Aspirin),¹⁶ FOURIER (TIMI 59) (Further Cardiovascular Outcomes Research with PCSK9 Inhibition in Patients with Elevated Risk),¹⁷ and DECLARE-TIMI 58 (Dapagliflozin Effect on Cardiovascular Events).¹⁸ Descriptions of specific trial populations are listed in the [Supplementary data online, Methods](#). Patients from each trial who consented to genetic analysis and passed quality control were included in the main analysis ($N = 68\,816$). Baseline characteristics of patients included from each trial have been previously reported.¹⁹ Study protocols were approved by the institutional review board or ethics committee at each participating site. All subjects provided written informed consent. Although data and study materials are not permitted to be made universally available, we encourage parties interested in collaboration to contact the corresponding author directly.

Genotyping and imputation

Methods for genotyping and imputation have been previously published.²⁰ Genotyping was performed using one of three arrays: Illumina Multi-Ethnic Genotyping Array (ENGAGE AF-TIMI 48, SAVOR-TIMI 53, and PEGASUS-TIMI 54), Affymetrix Biobank Array (SOLID-TIMI 52), and Infinium Global Array (FOURIER). Pre-imputation quality control was performed using PLINK v2.0,²¹ including filtering for sex mismatching, missingness >0.02 by sample and locus, Hardy–Weinberg equilibrium $P < 10^{-6}$, and a heterozygosity rate >3 SD from the mean. Imputation was performed on the Michigan Imputation server²² using the TOPMed Freeze5 reference panel,²³ excluding sites with an imputation quality score $r < 0.3$. To minimize population stratification bias, genetic ancestry was inferred using the 1000 Genomes phase 3 v5 reference panel¹⁹ and the ADMIXTURE tool²¹ with individuals assigned to one of five groups genetically similar to 1000 Genomes reference populations (African, Admixed American, European, South Asian, or East Asian) based on $>80\%$ genetic similarity to the respective reference population. Sub-ethnicity principal components (PCs) 1–5 were calculated using EIGENSOFT and used as an adjustment in all models containing the PRS.²⁴

Polygenic risk score

A PRS was calculated for each patient based on the previously published and validated 1.3-million single-nucleotide polymorphisms (SNPs) LDPred-auto model developed using summary statistics from the UK Biobank, All of Us, and MGB Biobank.²⁵ The raw genetic risk score was calculated for each patient using PLINK v2.0 by multiplying the imputed allelic dosage with a variant-specific weight and then summed across all variants. The PRS was assessed both continuously (per 1-SD) as well as categorically. The PRS

was categorized by comparing the top 20% (high) vs the middle 60% (intermediate) vs the bottom 20% (low), as well as by comparing the top 20% vs the bottom 80%. The same approach was used for other PRSs tested in comparison, which include a 1.3-million SNPs coronary artery disease (CAD) PRS from Patel *et al.*²⁶ and a 1.2 million stroke from the GIGA stroke consortium, Mishra *et al.*²⁷

Clinical endpoints

The endpoints of interest included a history of PAD at randomization and the MALE composite during trial follow-up, consisting of ALI, CLTI, major amputation, or peripheral revascularization, as well as the individual MALE components. The availability of each individual component endpoint varied by trial (see [Supplementary data online, Table S1](#)). Incident MI and ischaemic stroke were also examined in a supplementary analysis to assess the performance of the PAD-specific PRS for events across all vascular beds and in comparison with other disease-specific PRSs. Components of the MALE composite reported by investigators as adverse events, with independent review and adjudication by blinded physicians.

Statistical analysis

Logistic regression was used to calculate odds ratios (ORs) for prevalent PAD at the time of randomization in relation to the PRS with adjustment for the following: age, sex, genetic PCs 1–5, hypertension, hyperlipidaemia, diabetes mellitus, smoking status, history of heart failure, prior ischaemic stroke, chronic kidney disease, and trial. Additionally, we stratified individuals based on the number of clinical risk factors into three categories: 0–1, 2–3, and 4 or more risk factors. To control for population stratification, all models with the PRS included genetic PCs 1–5 as covariates, and the PRS was standardized within genetically inferred ancestry groups. While the primary models pooled data across ancestries with PC adjustment, ancestry-stratified analyses were also performed. We then assessed the association between PRS categories (low, intermediate, and high) and prevalent PAD within each risk factor category. For incident adverse limb events occurring during follow-up, time-to-event data were used to create Kaplan–Meier curves by genetic risk category. A Cox proportional hazards model was used to calculate the hazard ratios (HRs) for incident events across genetic risk categories, adjusted for the co-variables as the logistic regression model and also including trial-specific study treatment. A restricted cubic spline with 3 knots at quantiles, 0.1, 0.5, and 0.9 of the PAD PRS was applied to the same adjusted Cox model to derive individual predicted risk of MALE at 3 years, smoothed via LOESS. The proportional hazards assumption was assessed using Schoenfeld residuals and found to be satisfied. The start and end dates for the Cox regression analysis are shown in [Supplementary data online, Table S2](#). The association between the PAD PRS and PAD outcomes was assessed in key subgroups (prior PAD, smoking status, diabetes, and sex) to explore whether this relationship differs across these clinical factors. These subgroups were selected because they are established risk factors for PAD,^{4–6} and examining the association within these groups can provide insights into how genetic risk interacts with traditional risk factors. All reported P -values are two-sided with a threshold of 0.05 used for statistical significance. In addition, receiver operating characteristic curves were generated to evaluate the predictive performance of different models for PAD. The models included a base model consisting of age, sex, and trial, with additional models incorporating PAD PRS, clinical risk factors, and their combinations. Clinical risk factors were chosen following the clinical risk score developed by Duval *et al.*²⁸ and included hypertension, hyperlipidaemia, diabetes mellitus, smoking status, history of heart failure, prior ischaemic stroke, and chronic kidney disease. The area under the curve (AUC) was calculated for each model to quantify its discrimination ability. AUC estimates and confidence intervals (CIs) were derived from 1000 bootstrapped train–test splits and reported as medians with 25th and 75th percentiles. P -values were obtained empirically from the distribution of 1000 bootstrapped DeLong Z -statistics. Calibration curves were estimated nonparametrically using bootstrap over a sequence of

predicted values and smoothed via LOESS. In addition, pairwise continuous net reclassification index (NRI) was calculated between all models via bootstrap, reporting mean sensitivity and specificity improvements. All analyses were conducted in R v4.4.2.

Results

A total of 68 816 patients were included across the 6 trials, with a median follow-up of 2.6 (interquartile range 2.2–3.8) years. Baseline characteristics by genetic risk categories are presented in [Table 1](#). [Supplementary data online, Table S3](#) shows the number of patients in genetic risk categories across the included trials.

Association between PRS and PAD prevalence

A total of 5986 patients (8.7%) had known PAD at randomization. A higher PAD PRS was independently associated with a 15% greater odds of PAD at randomization after adjusting for clinical risk factors (OR_{adj} per 1-SD: 1.15 [95% CI: 1.12–1.18], $P < .0001$, [Supplementary data online, Figure S1](#)). When compared with subjects in the low PAD PRS category, patients with an intermediate PAD PRS had a 22% higher odds of PAD at randomization (OR_{adj} 1.22, 95% CI 1.13–1.32), while those with a high PAD PRS had a 43% higher odds (OR_{adj} 1.43, 95% CI 1.31–1.57; P -trend $< .0001$). The magnitude of risk for PAD conferred by a high PRS was as strong as many common clinical risk factors ([Figure 1](#)). The association between the PAD PRS and prevalent PAD was stronger among non-smokers (OR_{adj} per 1-SD: 1.17 [1.13, 1.21]) compared with current smokers (OR_{adj} per 1-SD: 1.09 [1.04, 1.16]; P -interaction = .0091). Additionally, our results showed an additive effect when integrating PRS with clinical risk factors ([Figure 2](#)). A model incorporating the PAD PRS alongside clinical risk factors statistically significantly, but not clinically meaningfully, improved the predictive accuracy, increasing the AUC from 0.66 (95% CI: 0.65–0.67) to 0.67 (95% CI: 0.66–0.68, Delong test P -value $< .0001$ [Supplementary data online, Figure S2A](#)). The

full model (age, sex, PAD PRS, and risk factors) demonstrated modest improvements in model discrimination across subgroups (see [Supplementary data online, Table S4](#)). There was no clear heterogeneity by age or clinical risk factor burden. The addition of the PAD PRS to the clinical risk factor model resulted in a continuous NRI of 0.11 (95% CI: 0.08–0.13), reflecting improved classification of both events (NRI+ 0.05 [0.02–0.08]) and non-events (NRI– 0.06 [0.05–0.07]) (see [Supplementary data online, Table S5](#)). Calibration plots demonstrated better alignment between predicted and observed risk, particularly at the high-end extreme, suggesting improved risk stratification despite limited gains in overall discrimination (see [Supplementary data online, Figure S2B](#)).

Association between PRS and incidence of MALE

During the follow-up period, 137 patients experienced ALI, 205 experienced CLTI, 183 underwent major amputations, and 649 underwent peripheral revascularizations. A total of 577 patients experienced MALE, including at least one of these events during follow-up (see [Supplementary data online, Table S1](#)). After adjustment for clinical risk factors, the PAD PRS was associated with a significant 30% increased risk of MALE (HR_{adj} per 1-SD: 1.30 [1.19–1.42], $P < .0001$). In terms of individual components, the adjusted HRs (95% CI) were: 1.18 (1.02–1.37) for CLTI, 1.26 (1.16–1.36) for peripheral revascularization, 1.08 (0.90–1.29) for ALI, and 1.15 (0.98–1.34) for major amputation. ([Figure 3](#)). All associations were consistent when the PAD PRS was analysed categorically ([Figure 3](#) and [Supplementary data online, Figure S3](#)). Notably, the association between the PAD PRS and MALE was consistent in individuals with and without PAD at baseline (MALE: no prior PAD: HR_{adj} per 1-SD 1.25 [95% CI: 1.11–1.42]; history of PAD: HR_{adj} per 1-SD 1.19 [95% CI: 1.04–1.36]; P -interaction = .60, [Supplementary data online, Figure S4](#)). However, the PAD PRS showed a stronger association with MALE in non-current smokers compared with current smokers (MALE: non-current smokers: HR_{adj} per 1-SD

Table 1 Baseline characteristics by categories of genetic risk

Variable	Low PRS N = 13 764	Intermediate PRS N = 41 286	High PRS N = 13 766	P-value
Age (years) $\pm$ SD	68.1 (8.8)	65.4 (8.7)	62.9 (8.9)	$< .0001$
Sex, male	9720 (70.6)	29 148 (70.6)	9340 (67.8)	$< .0001$
BMI, kg/m ²	29.5 (5.4)	30.3 (5.6)	30.8 (5.8)	$< .0001$
Diabetes mellitus	5775 (42.0)	22 245 (53.9)	8631 (62.7)	$< .0001$
Hypertension	11 362 (82.5)	34 362 (83.2)	11 622 (84.4)	.00011
Hyperlipidaemia	8102 (58.9)	26 457 (64.1)	9234 (67.1)	$< .0001$
Chronic kidney disease	2232 (16.2)	6990 (16.9)	2293 (16.7)	.15
Current smoker	1921 (14.0)	6978 (16.9)	2744 (19.9)	$< .0001$
Myocardial infarction	5523 (40.1)	20 988 (50.8)	8071 (58.6)	$< .0001$
Stroke	1571 (11.4)	4085 (9.9)	1285 (9.3)	$< .0001$
Peripheral artery disease	891 (6.5)	3607 (8.7)	1488 (10.8)	$< .0001$
Heart failure	3816 (27.7)	10 173 (24.6)	3199 (23.2)	$< .0001$

BMI, body mass index; PRS, polygenic risk score.

All values represent n (%) or mean $\pm$ SD unless otherwise indicated. P -values represent the χ^2 test for categorical variables and the one-way ANOVA for continuous variables.

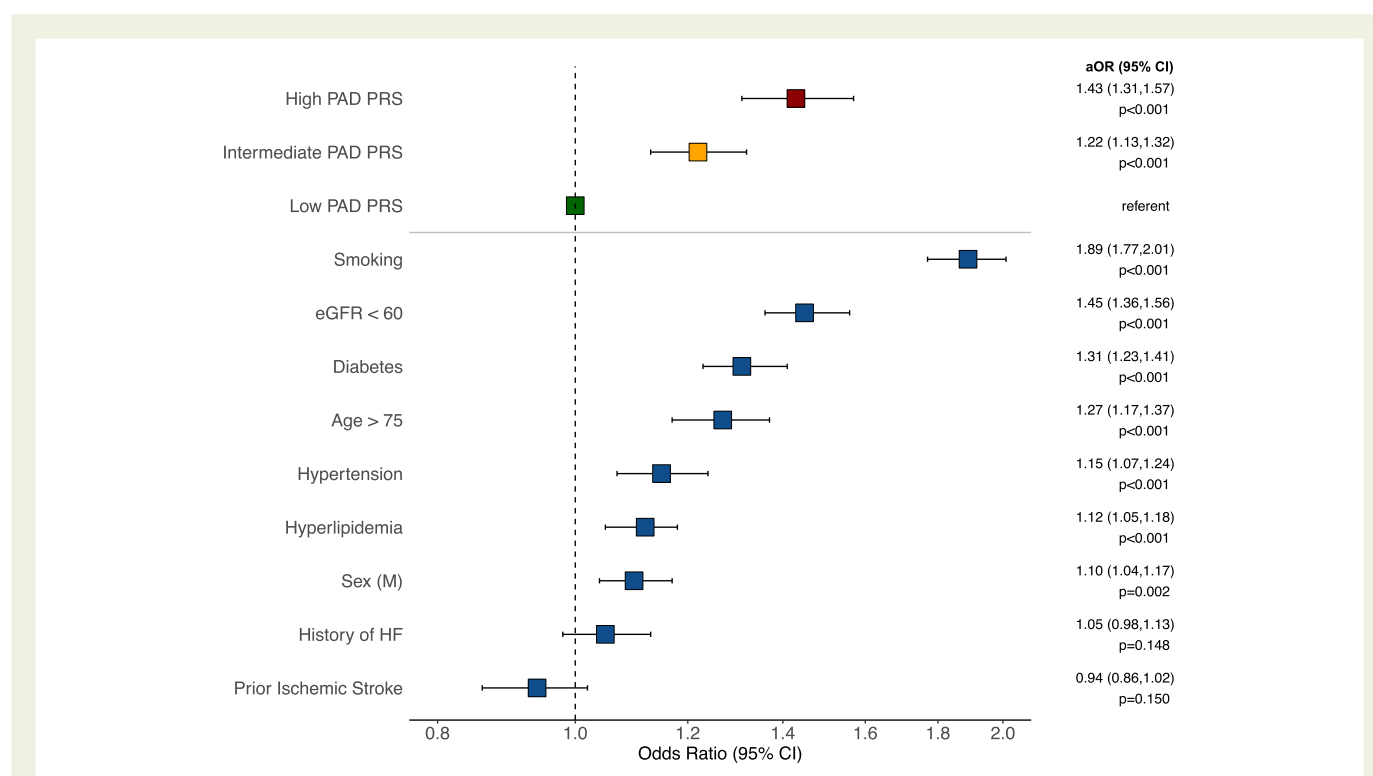


Figure 1 History of peripheral artery disease (PAD) at randomization by polygenic risk score (PRS) category—low (bottom 20%), intermediate (middle 60%), and high (top 20%)—and clinical risk factors. Odds ratios (ORs) are adjusted for trial and clinical covariates. eGFR, estimated glomerular filtration rate; PRS, polygenic risk score; PAD, peripheral artery disease; OR, odds ratio.

1.39 [95% CI: 1.25–1.55]; current smokers: HR_{adj} per 1-SD 1.09 [95% CI: 0.93–1.28]; P -interaction = .02, [Supplementary data online, Figure S5](#)). Moreover, the association was significantly stronger in individuals with diabetes mellitus compared with those without (MALE: no diabetes mellitus: HR_{adj} per 1-SD 1.06 [95% CI: 0.89–1.26]; with diabetes mellitus: HR_{adj} per 1-SD 1.40 [95% CI: 1.26–1.55]; P -interaction = 0.02, [Supplementary data online, Figure S6](#)). No differences were observed in relative risks between males: HR_{adj} 1.31 (95% CI: 1.18–1.45) and females: HR_{adj} 1.27 (95% CI: 1.06–1.52); P -interaction = 0.87 (see [Supplementary data online, Figures S7](#)). The full model demonstrated modest and consistent improvement in discrimination across all subgroups, with no clear evidence of heterogeneity in PRS performance (see [Supplementary data online, Table S6](#)). Adding the PAD PRS to a model with clinical risk factors resulted in a statistically significant but modest improvement in discrimination for 3-year MALE risk (AUC increased from 0.651 [95% CI: 0.624–0.678] to 0.662 [95% CI: 0.635–0.688]; DeLong test $P < .0001$; [Supplementary data online, Figure S8A](#)). Calibration plots showed improved alignment between predicted and observed risk, particularly at higher risk levels, suggesting improved stratification despite limited gains in overall discrimination (see [Supplementary data online, Figure S8B](#)).

Association between PRS and incidence of MI and ischaemic stroke

A total of 3134 MIs and 1402 ischaemic strokes occurred during follow-up. After adjustment for clinical risk factors, the PAD PRS was associated with a 27% increased risk of MI (HR_{adj} per 1-SD: 1.27 [1.22–1.32], $P < .0001$) and a 10% increased risk for ischaemic stroke during

follow-up after adjustment for clinical risk factors (HR_{adj} per 1-SD: 1.10 [1.04–1.16], $P = .0005$).

Comparison of PRSs for PAD, CAD, and stroke

We compared the associations of the PAD-specific PRS with the most updated CAD PRS by Patel *et al.*²⁶ and the GIGASTROKE stroke PRS²⁷ with PAD prevalence and incident MALE. (see [Supplementary data online, Tables S7](#) and [S8](#)). Although the CAD and stroke PRSs were significantly associated with PAD and limb events, the strength of these associations was attenuated compared with the PAD-specific PRS. There was a high degree of correlation between the PAD PRS and the CAD PRS (Spearman's $\rho = 0.84$), suggesting shared genetic architecture (see [Supplementary data online, Figure S9](#)). However, the CAD PRS was more strongly associated with CAD and MI, while the stroke PRS was most strongly associated with stroke risk (see [Supplementary data online, Tables S7](#) and [S8](#)).

Association between PAD PRS and PAD and MALE by groups genetically similar to reference populations

A total of 68 810 individuals were included in the genetic dataset, including 59 900 individuals genetically similar to the European reference population (8.8% with PAD at baseline), 1213 similar to the African reference population (9.6%), 4537 similar to the Admixed American reference population (11.7%), 1891 similar to the East Asian reference population (2.6%), and 1269 similar to the South Asian reference population (3.4%).

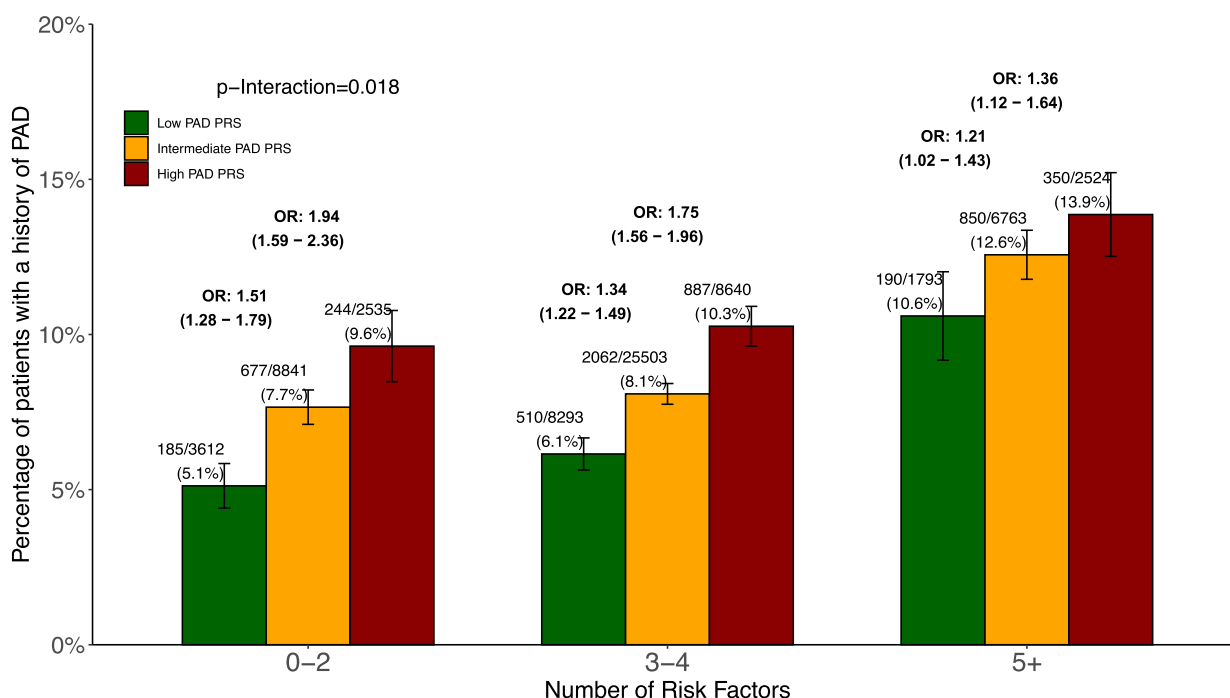


Figure 2 Gradient of effect on the association with PAD prevalence when integrating PRS and the number of clinical risk factors. Odds ratios for intermediate and high PRS are compared with the low PRS reference group. Risk factors include sex, chronic kidney disease, diabetes mellitus, smoking, hypertension, hyperlipidaemia, history of heart failure, and prior ischaemic stroke. PRS, polygenic risk score; PAD, peripheral artery disease; OR, odds ratio.

For PAD prevalence, the PAD PRS was significantly associated with PAD among individuals genetically similar to the European population (OR_{adj} per 1-SD: 1.18 [95% CI: 1.14–1.21]) and those similar to the South Asian population (OR_{adj} : 1.42 [1.03–1.95]). No significant associations were observed in those similar to the African (OR_{adj} : 0.93 [0.76–1.13]), Admixed American (OR : 0.99 [0.89–1.09]), or East Asian populations (OR_{adj} : 1.11 [0.82–1.51]) (see [Supplementary data online, Table S9](#)).

For incident MALE, the PAD PRS was significantly associated with risk in individuals genetically similar to the European (HR_{adj} per 1-SD: 1.30 [1.19–1.43]). Associations were directionally consistent in individuals similar to African (HR_{adj} : 1.05 [0.56–1.99]), Admixed American populations (HR_{adj} : 1.34 [0.84–2.15]), and South Asian (HR_{adj} : 1.41 [0.29–6.77]). No HR could be calculated for individuals similar to the East Asian reference populations as only 1 event occurred during the follow-up (see [Supplementary data online, Table S10](#)).

Discussion

Leveraging data from almost 70 000 genotyped individuals across six cardiovascular outcomes trials, we demonstrated that a PAD PRS is associated with an increased risk of PAD and with the incidence of MALE, beyond clinical risk factors, in a patient population spanning the spectrum of cardiometabolic disease ([Structured Graphical Abstract](#)).

Our findings build upon previous analyses, which have identified genetic risk loci specific to PAD.^{9,12,25} These prior studies have also suggested that incorporating a PRS for PAD into a clinical risk model could enhance the identification of existing yet undiagnosed PAD.¹² In our analysis, we further expand these findings by applying this PAD

PRS in a patient population largely with established cardiometabolic disease and demonstrate that patients at the top third of genetic risk have a >1.5-fold increased odds of prevalent PAD after more comprehensive adjustment for clinical risk factors. The magnitude of risk for PAD conferred by a high PRS was as strong as many common clinical risk factors ([Figure 1](#)). Furthermore, we observed the ability of the PAD PRS to identify a gradient of PAD prevalence in patients across the spectrum of clinical risk factor burden.

Moreover, we demonstrated that the PAD PRS was not only associated with increased odds of prevalent PAD but also with a significantly elevated risk of incident adverse limb events. Individuals in the top one-third of the PRS had approximately a two-fold increased hazard of incident MALE after adjustment for clinical risk factors. Among the components of MALE, the PRS showed the strongest associations with CLTI and peripheral revascularization, suggesting that the PAD PRS may primarily capture atherosclerotic or inflammatory pathways, which align more closely with chronic ischaemic phenotypes such as CLTI.^{29–32} In contrast, the adjusted HR for ALI was closer to 1, which may reflect the heterogeneous nature of ALI, often driven by acute embolic or thrombotic events.^{29–32} The weaker association between the PAD PRS and major amputation as compared with peripheral revascularization may be explained by multiple factors. Beyond obstructive PAD, microvascular disease might contribute to an elevated amputation risk in specific high-risk subgroups, such as patients with chronic renal insufficiency and diabetes.^{5,33,34} Additionally, decisions regarding amputation incorporate clinical judgment regarding limb salvageability.

Notably, the association between the PAD PRS and MALE was consistent regardless of PAD status at baseline. However, the association

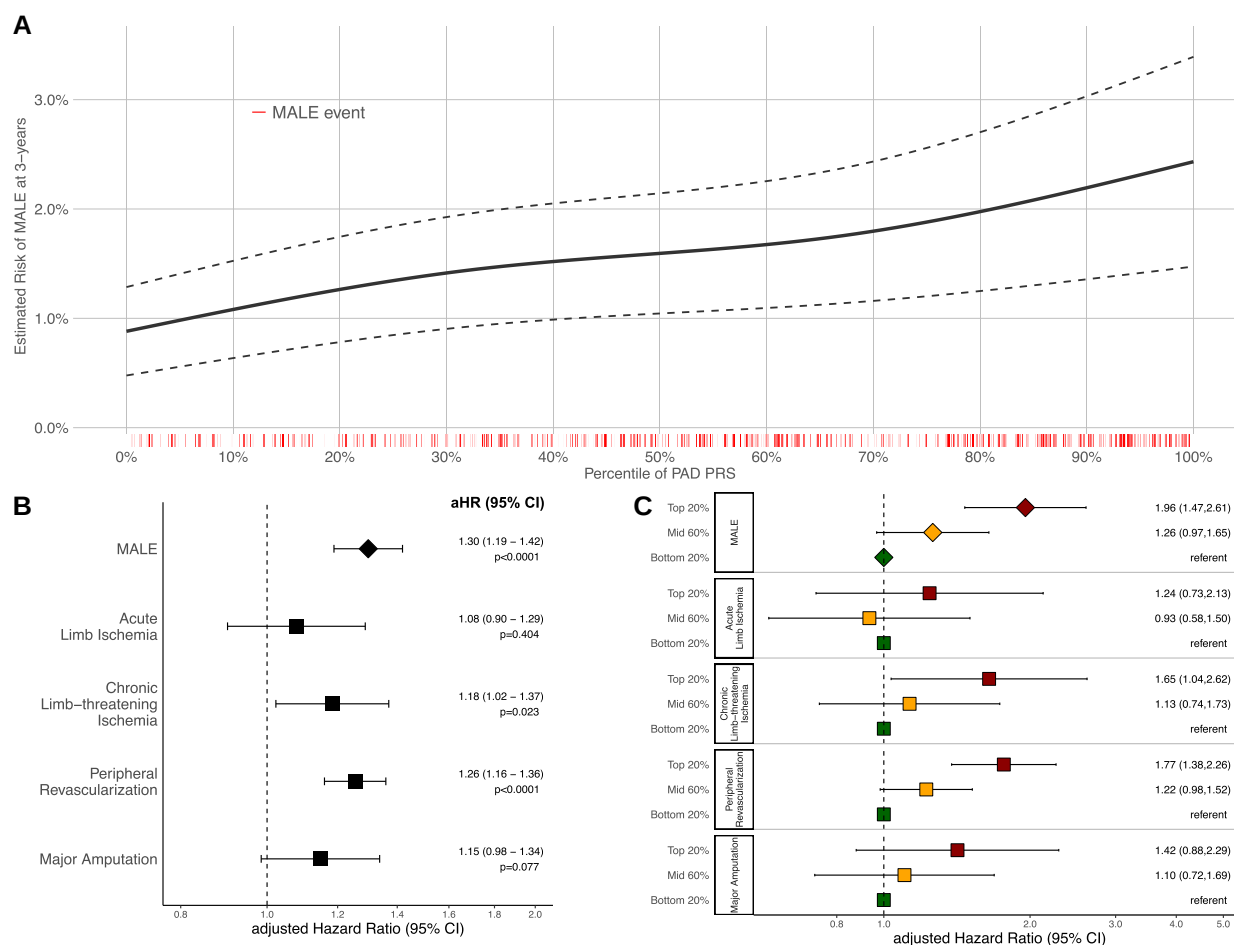


Figure 3 (A) Adjusted hazard ratio per 1-SD increase in PAD PRS for incident major adverse limb events (MALE). Density plot above x-axis represents the distribution of MALE events across PAD PRS. (B) Forest plot of adjusted hazard ratios per 1-SD increase in PAD PRS for individual MALE components. (C) Forest plot of adjusted hazard ratios for MALE components stratified by PRS category: low (bottom 20%), intermediate (middle 60%), and high (top 20%). All models are adjusted for age, sex, genetic ethnicity, hypertension, hyperlipidaemia, diabetes mellitus, smoking status, history of heart failure, prior ischaemic stroke, chronic kidney disease, and trial-specific treatment arm. CI, confidence interval; PRS, polygenic risk score; HR, hazard ratio; PAD, peripheral artery disease.

was more pronounced among non-smokers, suggesting potential effect modification by smoking. In the presence of a strong environmental risk factor like smoking, genetic predisposition may have a relatively smaller influence on MALE risk. Conversely, the association between the PAD PRS and MALE was stronger in individuals with diabetes mellitus, indicating that genetic susceptibility may be more relevant in the context of metabolic stressors such as diabetes. Given the limited sample sizes in these subgroups, our findings could be due to the play of chance and should be considered hypothesis-generating. Future studies with larger cohorts are necessary to determine the potential interplay between genetics, lifestyle, and comorbid diseases in predicting PAD and limb events.

Many of the loci identified in the PAD GWAS are linked to established risk factors for PAD, including lipids, type 2 diabetes, smoking, thrombosis, and hypertension.^{9,12,25} Notably, several SNPs intricately participate in well-established lipid pathways, such as the LDL receptor, lipoprotein lipase, and lipoprotein(a), establishing a direct connection to the recognized impact of lipid-lowering therapies on PAD risk. Additionally, a DNA gene variant in the *CHRNA3* locus exhibits a robust correlation with variants predictive of nicotine dependence, underscoring

its pivotal role in influencing PAD risk, especially among individuals who smoke. As our genetic analyses of PAD advance in both sample size and scope, these findings pave the way for uncovering additional therapeutic targets that hold the potential to uniquely benefit patients by addressing the intricate pathways involved in adverse limb events.^{29,30}

Given the high prevalence of CAD in patients with PAD,^{4,5} we also evaluated the performance of the PRS for incident MI. While the PAD PRS was associated with an increased risk of MI, this association was modest compared with that observed for adverse limb events, suggesting some overlap in the underlying atherosclerotic mechanisms across vascular beds. In contrast, no association was observed between the PAD PRS and ischaemic stroke. These findings highlight the vascular bed-specific performance of PRSs and suggest that, despite shared atherosclerotic biology, distinct clinical and genetic factors likely drive differential risk across arterial territories.

We hypothesize that the stronger association of the PAD PRS with adverse limb events reflects the inclusion of SNPs linked to pathways particularly relevant to peripheral atherosclerosis, thrombosis, and vascular remodelling. While some variants overlap with CAD and stroke

PRSs, the PAD PRS includes loci such as F5 (rs6025), MMP3 (rs566125), and IL6 (rs4722172), which are implicated in pro-thrombotic and inflammatory processes relevant to MALE.^{9,12,25} Other variants, including CHRNA3 and CDKN2B-AS1/9p21, may influence plaque instability and vascular remodelling in the peripheral circulation.^{35,36} Similar to how the CAD PRS associates with non-invasive measures of coronary plaque burden (e.g. coronary computed tomography angiography),³⁷ the PAD PRS may identify individuals with greater susceptibility to peripheral atherosclerosis—placing them at elevated risk for PAD and limb events. These features likely enhance the PAD PRS's sensitivity to limb-specific ischaemic risk, distinguishing it from PRSs for CAD or stroke.

Limitations

The study comprised participants from six randomized clinical trial populations, so it may not be generalizable to the broader population. Furthermore, our analysis focused mainly on subjects of European ethnicity, and the extent to which our results apply to other ethnicities remains uncertain. Additionally, MALE were investigator-reported and not centrally adjudicated events. The included studies did not include the ankle-brachial index at baseline. Moreover, the assessment of MALE was not uniform across the included trials. A key limitation of this study is the possibility that PAD may have been undiagnosed at baseline in patients with no prior history of PAD, with the diagnosis only being established after the occurrence of MALE during follow-up. Furthermore, the relatively short follow-up period (2–3 years) may limit the ability to detect the full magnitude of the association with the development of MALE, particularly in patients without prior PAD. Future studies should consider more comprehensive diagnostic evaluations and aim to better distinguish the causes of MALE, including PAD and other vascular conditions.

Clinical implications

Although the PAD PRS identifies individuals at higher risk of developing PAD and subsequent limb events independent of clinical factors, the improvement in discrimination when added to clinical risk factors was modest. Thus, at present, the immediate clinical utility of incorporating the PAD PRS into routine risk assessment appears limited. Future research should evaluate whether refinements in PRS development, or integration with other biomarkers, could improve risk stratification.

Conclusions

In a broad spectrum of patients with cardiometabolic disease, we demonstrated that the PAD PRS is associated with an increased risk of PAD and the incidence of MALE beyond clinical risk factors; the improvement in discrimination was statistically significant but clinically modest.

Supplementary data

Supplementary data are available at [European Heart Journal](https://www.heartjnl.com) online.

Declarations

Disclosure of Interest

S.A.S. reports grants from the German Research Foundation (DFG) (GZ: SA 4598/2-1) during the conduct of the study and is currently employed by Novartis outside the submitted work.

S. M. P. reports consultant/advisory fees from Janssen.

G.E.M.M. reports no conflict of interest.

F.K.K. reports no conflict of interest.

R.P.G. reports clinical trials/research support from Amgen, Anthos Therapeutics, Daiichi Sankyo, and Ionis. He has received honoraria for lectures/CME programs from Amgen, Big Sky Cardiology, CADECI, Daiichi Sankyo, Dr. Reddy's Laboratories, Medical Education Resources (MER), Menarini, and SUMMEET. He also serves as a consultant for AstraZeneca, Beckman Coulter, Daiichi Sankyo, Inari, Inventiva, Perosphere, PhaseBio Pharmaceuticals, Samsung, and Sanofi Aventis.

S.D.W. is currently an employee at AstraZeneca, and reports research grants from Amgen, AstraZeneca, Janssen, Merck, and Pfizer during the conduct of the study. He reports consulting fees from Icon, Novo Nordisk, and Verian. He reports speaking honoraria from Harvard Medical School. Spouse, Dr Caroline Fox, is a former employee of Merck, Flagship Labs, Vertex and a current employee of Pfizer.

B.M.S. reports institutional research support to Brigham and Women's Hospital from Amgen, Better Therapeutics, Boehringer Ingelheim, Merck, NovoNordisk, Pfizer, and Verve Therapeutics. Consulting fees from AbbVie (DSMB), Amgen, AstraZeneca (DSMB), Bayer, Boehringer Ingelheim (DSMB), Elsevier Practice Update Cardiology, Hanmi (DSMB), Lexeo (DSMB), NovoNordisk, Verve Therapeutics, and equity in Health [at] Scale.

M.L.O.D. reports grants received by The TIMI Study Group through the institution from Amgen, AstraZeneca, Novartis, Merck, and Janssen Medicines Company. She has received consulting and/or DSMB fees from Amgen, Novartis, Janssen, AstraZeneca, and CRICO, as well as honoraria from Medscape Cardiology.

C.P.C. reports research grants from: Amgen, Better Therapeutics, Boehringer-Ingelheim (BI), Novo Nordisk, and salary support from Colorado Prevention Center (CPC) Clinical Research, which gets research grant support from Amgen, Bayer, Cleerly, Esperion, Lexicon, Silence, and consulting fees from Amryt/Chiesi, Amgen, Ascendia, Biogen, BI, BMS, CSL Behring, Eli Lilly, Janssen, Lexicon, Milestone, Pfizer, and Rhoshan. He also serves on Data and Safety Monitoring Boards for the Areteia, Novo Nordisk, ROMTherapy, Inc., and the Veterans Administration.

D.L.B. discloses the following relationships—Advisory Board: Angiowave, Antlia Bioscience, Bayer, Boehringer Ingelheim, CellProthera, Cereno Scientific, E-Star Biotech, High Enroll, Janssen, Level Ex, McKinsey, Medscape Cardiology, Merck, NirvaMed, Novo Nordisk, Repair Biotechnologies, Stasys, and Tourmaline Bio; Board of Directors: American Heart Association New York City, Angiowave (stock options), Bristol Myers Squibb (stock), DRS.LINQ (stock options), and High Enroll (stock); Consultant: Alnylam, Altimmune, Broadview Ventures, Corcept Therapeutics, Corsera, GlaxoSmithKline, Hims, SERB, SFJ, Summa Therapeutics, and Worldwide Clinical Trials; Data Monitoring Committees: Acesion Pharma, Assistance Publique-Hôpitaux de Paris, Baim Institute for Clinical Research, Boston Scientific (Chair, PEITHO trial), Cleveland Clinic, Contego Medical (Chair, PERFORMANCE 2), Duke Clinical Research Institute, Mayo Clinic, Mount Sinai School of Medicine (for the ABILITY-DM trial, funded by Concept Medical; for ALLAY-HF, funded by Alleviant Medical), Novartis, and Population Health Research Institute; Rutgers University (for the NIH-funded MIINT Trial); Honoraria: American College of Cardiology (Senior Associate Editor, Clinical Trials and News, ACC.org; Chair, ACC Accreditation Oversight Committee), Arnold and Porter law firm (work related to Sanofi/Bristol-Myers Squibb clopidogrel litigation), Baim Institute for Clinical Research (AEGIS-II executive committee funded by CSL

Behring), Belvoir Publications (Editor in Chief, Harvard Heart Letter), Canadian Medical and Surgical Knowledge Translation Research Group (clinical trial steering committees), CSL Behring (AHA lecture), Duke Clinical Research Institute, Engage Health Media, HMP Global (Editor in Chief, Journal of Invasive Cardiology), Medtelligence/ReachMD (CME steering committees), MJH Life Sciences, Oakstone CME (Course Director, Comprehensive Review of Interventional Cardiology), Philips (Becker's Webinar on AI), Population Health Research Institute, WebMD (CME steering committees), and Wiley (steering committee); Other: Clinical Cardiology (Deputy Editor); Progress in Cardiovascular Diseases (Deputy Editor); Patent: Sotagliflozin (named on a patent for sotagliflozin assigned to Brigham and Women's Hospital who assigned to Lexicon; neither I nor Brigham and Women's Hospital receive any income from this patent); Research Funding: Abbott, Acesion Pharma, Afimmune, Alnylam, Amarin, Amgen, AstraZeneca, Atricle, Bayer, Boehringer Ingelheim, Boston Scientific, CellProthera, Cereno Scientific, Chiesi, Cleerly, CSL Behring, Faraday Pharmaceuticals, Fractyl, Idorsia, Janssen, Javelin, Lexicon, Lilly, Medtronic, Merck, MiRUS, Moderna, Novartis, Novo Nordisk, Pfizer, PhaseBio, Regeneron, Reid Hoffman Foundation, Roche, Sanofi, Stasys, and 89Bio; Royalties: Elsevier (Editor, Braunwald's Heart Disease); and Site Co-Investigator: Cleerly.

E.M.A. reports grants from Daiichi Sankyo during the conduct of the study and being a member of the TIMI Study Group.

E.B. reports research grants through his institution from AstraZeneca, Daiichi Sankyo, Merck, and Novartis, and consultancies with Amgen, Bristol Myers Squibb, Boehringer Ingelheim/Lilly, Cardurion, Edgewise, and Verve.

P.T.E. reports grants from Bayer, IBM Research, Novo Nordisk, Pfizer, and Bristol Myers Squibb, and has served on advisory boards or consulted for Bayer, MyoKardia, and Novartis.

M.P.B. is the Executive Director of CPC, a nonprofit academic research organization affiliated with the University of Colorado, which receives or has received research grant/consulting funding between August 2021 and present from: Abbott Laboratories, Agios Pharmaceuticals, Alexion Pharma, Alnylam Pharmaceuticals, Amgen, Angionetics, Anthos Therapeutics, ARCA Biopharma, Array BioPharma, AstraZeneca and Affiliates, Atentiv, Audentes Therapeutics, Bayer and Affiliates, Beth Israel Deaconess Medical Center, Better Therapeutics, Boston Clinical Research Institute, Bristol-Meyers Squibb Company, Cambrian Biopharma, Cardiol Therapeutics CellResearch, Cleerly, Cook Regentec, CSL Behring, Eidos Therapeutics, EP Trading, EPG Communication Holdings, Epizon Pharma, Esperion Therapeutics, Everly Well, Exicon Consulting, Faraday Pharmaceuticals, Foresee Pharmaceuticals, Fortress Biotech, HDL Therapeutics, HeartFlow, Hummingbird Bioscience, Insmid, Ionis Pharmaceuticals, IQVIA, Janssen and Affiliates, Kowa Research Institute, Kyushu University, Lexicon Pharmaceuticals, Medimmune, Medpace, Merck & Affiliates, Nectero Medical, Novartis Pharmaceuticals, Novo Nordisk, Osiris Therapeutics, Pfizer, PhaseBio Pharmaceuticals, Pharmaceutical Product Development LP, Prairie Education and Research Cooperative, Prothena Biosciences Limited, Regeneron Pharmaceuticals, Inc., Regio Biosciences, Saint Luke's Hospital of Kansas City, Sanifit Therapeutics S.A., Sanofi-Aventis Groupe, Silence Therapeutics, Smith & Nephew, Stanford Center for Clinical Research, Stealth BioTherapeutics, State of Colorado Cancer, Cardiovascular, and Pulmonary Disease Grant, Brigham & Women's Hospital, Feinstein Institutes for Medical Research, Thrombosis Research Institute, University of Colorado, University of Pittsburgh,

VarmX, Virta Health Corporation, Worldwide Clinical Trials, WraSer, and Yale Cardiovascular Research Group.

M.S.S. reports research grant support through Brigham and Women's Hospital from: Abbott; Amgen; Anthos Therapeutics, Inc.; AstraZeneca; Boehringer Ingelheim; Daiichi-Sankyo; Ionis; Merck; Novartis; Pfizer; Sagmos Therapeutics; and Verve Therapeutics, and consulting for: Amgen; AMPEL BioSolutions; Anthos Therapeutics, Inc.; AstraZeneca; Beren Therapeutics; Boehringer Ingelheim; Dr. Reddy's Laboratories; Fibrogen; Merck; Moderna; Novo Nordisk; Precision BioSciences; and Silence Therapeutics.

N.A.M. reports clinical trials/research support from the National Institutes of Health and involvement in clinical trials with Amgen, Ionis, Pfizer, Novartis, and AstraZeneca. Consultant from Beckman Coulter, Cleerly, Inc. (CPC Clinical Research), Janssen, and Viz.ai. Honoraria for Lecture/CME Programs from Amgen, Illumina, and Medical Education Speakers Network (MESN).

C.T.R. reports research grant through institution: Anthos, AstraZeneca, Daiichi Sankyo, Janssen, and Novartis; Honoraria for scientific advisory boards and consulting: Anthos, Bayer, Bristol Myers Squibb, Daiichi Sankyo, Janssen, and Pfizer.

S.A.S., S.M.P., G.E.M.M., F.K.K., R.P.G., S.D.W., B.M.S., M.L.O.D., E.M.A., E.B., M.S.S., N.A.M., and C.T.R. are members of the TIMI (Thrombolysis in Myocardial Infarction) Study Group, which has received institutional research grant support through Brigham and Women's Hospital from Abbott, Abiomed, Inc., Amgen, Anthos Therapeutics, ARCA Biopharma, Inc., AstraZeneca, Boehringer Ingelheim, Daiichi-Sankyo, Ionis Pharmaceuticals, Inc., Janssen Research and Development, LLC, MedImmune, Merck, Novartis, Pfizer, Regeneron Pharmaceuticals, Inc., Roche, Sagmos Therapeutics, Inc., Softcell Medical Limited, The Medicines Company, Verve Therapeutics, Inc., and Zora Biosciences.

Data Availability

Although data and study material are not permitted to be made universally available, we encourage parties interested in collaboration to contact the corresponding author directly.

Funding

S.A.S. is supported by a postdoctoral grant from the German Research Foundation (Deutsche Forschungsgemeinschaft (DFG)_GZ: SA 4598/2-1). S.M.P. is supported by a K12TR004381 award through Harvard Catalyst | The Harvard Clinical and Translational Science Center (National Center for Advancing Translational Sciences, National Institutes of Health).

Ethical Approval

Study protocols were approved by the institutional review board or ethics committee at each participating site. All subjects provided written informed consent.

Pre-registered Clinical Trial Number

A genetic cohort analysis was performed by pooling individual patient-level data across 6 clinical trials across the spectrum of cardiometabolic disease: ENGAGE AF-TIMI 48,¹¹ SOLID-TIMI 52,¹² SAVOR-TIMI 53,¹³ PEGASUS-TIMI 54,¹⁴ FOURIER (TIMI 59),¹⁵ and DECLARE-TIMI 58.¹⁶ Description of specific trial populations is listed in the [Supplementary data online, Methods](#).

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